

EDUCATION

Heritage High School, Frisco, TX

Class of 2027

GPA 4.00 UW / 5.361 W · Rank 11/481 · ACT 36 · Nine AP scores of 5 · AP Scholar with Distinction

Senior coursework: AP Calculus BC, AP Physics 2, Computer Science III, Mobile App Programming

UT Dallas NLP with Python Workshop, Summer 2025 — Implemented rule-based text simplification using dependency parsing, frequency-based lexical substitution, clause splitting, and passive-to-active transformation.

RESEARCH

PAPER 1 · COMPUTATIONAL LINGUISTICS / CLINICAL ML

Predicting Cognitive Function Using Transformer-Derived Speech Representations and Longitudinal Coherence Features [↗] May 2025–PresentFirst Author · Mentor: Joe Xiao (PhD student) · CRediT: conceptualization, methodology, software, formal analysis, validation, original draft
medRxiv preprint (Aug. 2026) · Under review, *Frontiers in Aging Neuroscience* · [Public code](#) · [26-entry research log](#)

QUESTION	Can transcribed speech plus the clinical history already on file forecast a patient's next-visit MMSE (0–30)? Framed as longitudinal, patient-level supervised regression — forecasting, where most prior speech work does single-visit classification.
DATA	DementiaBank Pitt Corpus (controlled access): 126 patients (69 dementia, 57 control), 2–5 visits each → 173 prediction samples , each mapping 1–4 prior visits to a later-visit MMSE.
FEATURES	Designed a 53-dimensional feature vector: 12 per-visit features — Sentence-BERT discourse coherence (adjacent-sentence cosine similarity, global coherence, semantic slope, topic drift, embedding variance) plus spaCy lexical/syntactic measures (filler rate, 50-token MATTR, inter-sentence repetition, POS-based fragment ratio, named-entity rate) — with 8 visit-to-visit Δ features, 4 temporal features, 20 PCA components of 384-d SBERT embeddings, and 9 clinical/demographic covariates.
MODEL	LightGBM, chosen for small mixed-type tabular data: RobustScaler → correlation pruning ($ r > 0.8$, clinical variables kept) → permutation-importance selection (top 15) → quantile binning → 3× Gaussian augmentation (2% of feature SD) tagged by parent patient → Optuna TPE over 10 hyperparameters, 200 trials, maximizing mean patient-grouped R^2 .
VALIDATION	Diagnosed patient-identity leakage : folds split by visit, so a model could score by recognizing a person. Rebuilt evaluation as 5-fold GroupKFold on patient ID , locking augmented copies to their parent's fold and scoring only out-of-fold predictions on original samples ; scores fell, as they should once leaked identity is removed.
DEBUGGING	Traced why entry MMSE — the expected strongest predictor — contributed almost nothing to two silent data-join bugs (a renamed column; a string-vs-integer patient-ID mismatch) that imputed one value for every patient; the fix was the project's largest single improvement.
RESULTS	R^2 0.840 ± 0.015 · RMSE 2.75 · MAE 2.02 MMSE points · Pearson r 0.917 — error comparable to MMSE inter-rater variability. Per-fold permutation importance (10 repeats): entry MMSE, baseline diagnosis and Blessed score lead; SBERT PC1 ranks third.
ABLATION	Controlled feature-source ablation (fixed untuned LightGBM, same folds): linguistic-only R^2 0.118 , SBERT-only 0.458 , clinical-only 0.716 , clinical + SBERT 0.759 . Clinical history dominates; SBERT beats handcrafted features and adds directional, complementary signal (fold SDs overlap). Dropping Δ features changed nothing (0.740), overturning an earlier claim.
INTEGRITY & LIMITS	Audited the manuscript line by line against the code , removing six coreference/entity features the model never used. Limits: non-nested selection and tuning (CV may be optimistic); one cohort, no external validation; over-prediction below MMSE 10.

PAPER 2 · X-STEP RESEARCH PAPER · BIOMEDICAL SENSING

A Four-Site Plantar-Pressure Insole: Transducer Error Budget, Feature Equivariance, and Sparse-Versus-Dense Agreement in Overground Walking [↗] 2025–Present2nd of 5 authors · ML lead · MIT CSAIL first author; [UC Santa Cruz AIEA Lab](#) (Prof. Leilani Gilpin) · Under review, IEEE-EMBS IECBES 2026

FRAMING	Evaluates the team's four-FSR402 insole (MET1, MET2, MET5, heel; ESP32, 25 Hz, BLE) as a measurement instrument, not a clinical classifier . My contributions: the ML pipeline and the sparse-vs-dense correlation study; the hardware was team work.
ML PIPELINE	Designed the feature map from rolling 4 s windows (100 samples × 8 channels) to 59 features — per-site peak, mean, coefficient of variation and pressure–time integral; window-level cadence, stance ratio, anterior load share, aggregate peak and total PTI — feeding a 540-parameter logistic-regression head (531 multiply–accumulates; ≈0.23 ms per update at 25 Hz).
CORRELATION STUDY	Designed a sparse-vs-dense study on an independent 32-cell insole archive (149 recordings, 15 adults, 298 foot-takes), comparing the cell nearest each X-Step site with its anatomical cluster maximum. Shape survived (median r 0.72–0.99), magnitude did not (peak-to-peak r 0.43–0.74); four-site center of pressure tracked anteroposterior (r 0.81) but not mediolateral motion (r 0.01) — a geometric limit of a layout with no medial–lateral pair. Reported as descriptive (non-independent foot-takes, no p-values); Pearson r certifies shape, not magnitude.
EVALUATION	Quantified subject-identity leakage on a synthetic 24-subject, 9-class cohort: macro-F1 0.885 subject-grouped vs 0.931 ungrouped (majority baseline 0.040) — a transfer gap larger than the spread across all classifiers tested. Peak-only features matched the full map (0.886), read as peak-separable synthetic labels, not proof the full map is needed.
SCOPE	No walking data from X-Step wearers yet; findings are engineering-level, with no ulcer-prediction or clinical-benefit claim.

COMPUTING EDUCATION

KidCode — *Founder & Lead Instructor* · Frisco, TX

Jun. 2024–Present

124 students · 3 annual cohorts (35 / 53 / 36 new) · 10 Frisco neighborhoods · **120-class-day curriculum** · **20 projects** · median 10 per student (max 18)

- CURRICULUM** Designed a three-year, 120-class-day, 20-project (6 + 7 + 7) curriculum: Year 1 Python foundations; Year 2 assert-based unit tests, file I/O, JSON, exception handling, classes and inheritance, libraries; Year 3 spaCy NLP — tokenization, lemmatization, sentence segmentation, POS tagging, NER, Matcher rule patterns, and a rule-based chatbot.
- SPIRAL DESIGN** Earlier projects return at a higher level: Mad Libs becomes Auto Mad Libs with spaCy choosing the blanks; the trivia quiz returns JSON-backed with error handling; Year 2 ends by breaking students' own text-cleaning code, which Year 3 opens by testing against spaCy.
- ASSESSMENT** Wrote 13 curriculum-aligned pre/post assessment pairs (same skills and formats, different code). Paired results, **n = 104**: mean **41.7% → 58.8% (+17.1 pp)**; **99/104** improved, 4 unchanged, 1 declined. Observed change only; no comparison group.
- STUDENT WORK** Students completed **1,214** working programs across the 20 projects, from Python basics in Year 1 to spaCy NLP in Year 3 (all **572** Year 3 programs use spaCy).

Ayusha Nilayam — *Programming & Math Instructor* · Hyderabad, India (remote)

Nov. 2025–Jul. 2026

- SCOPE** Taught 38 weekly Zoom sessions of Python and math to **11 children** at a children's home, plus two weeks in person.
- CONTENT** Python from `print()` through conditionals, loops and functions to independent projects (calculator, number-guessing game); math from equations and inequalities through the unit circle, and statistics through standard deviation.
- CONSTRAINTS** Adapted to one shared laptop, unreliable internet and a Telugu–English language gap with frequent checks for understanding and student explain-backs — practices later carried back to KidCode.

ENGINEERING

X-STEP STARTUP · PRODUCT BUILT ON THE PAPER 2 INSOLE

X-Step — *Co-Founder & ML/Mobile Lead* · student startup team · working prototype, no wearer data yet

2025–Present

- PRODUCT** A plantar-pressure insole for people at risk of diabetic foot problems, built by the team: four FSR402 sensors per foot on an ESP32 at 25 Hz, streaming sequence-numbered 28-byte BLE frames to a React Native (Expo) app with live pressure, trends, alerts and a clinician view.
- MY ROLE** Designed the ML pipeline that turns the stream into alerts: rolling 4 s windows → 59 pressure and gait features → 540-parameter logistic-regression head, fast enough to run between 40 ms samples.

ShopAI — *Creator & Sole Developer* · live on the Apple App Store

2025–Present

One grocery search across nine retailers, plus multi-store trip planning · solo rebuild of a team hackathon prototype · Architecture write-up

- ARCHITECTURE** React Native/Expo client over a standalone Express/TypeScript backend on **AWS EC2 behind Nginx**; the client holds no search logic — every search, plan and warm-up check runs server-side through a four-layer, one-direction pipeline.
- ADAPTERS** Designed a **retailer-adapter abstraction** normalizing **nine chains** (seven with live data) into one schema via four integration strategies: OAuth2 client-credentials REST (Kroger, Harris Teeter), anonymous-session GraphQL (Aldi, Sprouts, Publix via Instacart), headless Playwright (Trader Joe's), server-rendered HTML (Whole Foods). Target excluded (terms of service).
- SEARCH PATH** Query correction → **concurrent fan-out** racing a shared response-time budget → per-store food/relevance filtering and ranking → explicit ok / error / unavailable / pending status per store. A per-ZIP **state machine** (starting → warming → ready) pre-warms OAuth tokens, cookies and the browser session off the request path.
- PLANNER** A **deterministic optimizer** exhaustively scores all $2^7 - 1 = 127$ non-empty store subsets, so “cheapest” is exact; cheapest, fastest and fewest-stops are single-field sorts; only “balanced” is weighted multi-criteria. **Zero** LLM calls in search or planning.
- PRODUCTION** Debugged a Node.js `setTimeout` overflow in a 30-day location cache that stalled the event loop (≈ 1.5 s → 27 s searches) by capping the sweep interval; shipped by feature-flagging off unfinished meal planning, AI assistant, OCR and barcode scanning.

RECOGNITION

- STARTUP** X-Step startup team: Texas A&M Ideas Challenge — **Finalist**, 2026 · Diamond Challenge — **Semifinalist**, 2025
- HACKATHONS** ShopSmart team prototype (2025): TechLit Hackathon — **1st** · PeddieHacks — **Innovation Prize** · Hack4Humanity — **3rd**

TECHNICAL SKILLS

- LANGUAGES** Python, TypeScript/JavaScript, Java, SQL
- ML & NLP** LightGBM, scikit-learn, Optuna (TPE), PyTorch, PCA, permutation importance, GroupKFold, ablation design; spaCy (tokenization, POS, NER, Matcher, dependency parsing), Sentence-BERT; windowed sensor-feature extraction
- SYSTEMS** React Native/Expo, Node.js/Express, REST, GraphQL, OAuth2, Playwright, AWS EC2, Nginx